

MATH 180 Strategies of Problem Solving

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October 6, 2026

Last time

Last time

- a story about Gauss

Last time

- a story about Gauss
- problem solving strategies

Last time

- a story about Gauss
- problem solving strategies
- math competition problems

Play the game

Activity

Play the game

Activity

- remind yourself how to play

Play the game

Activity

- remind yourself how to play
- play a some games of Connect 4 with a friend

Play the game

Activity

- remind yourself how to play
- play a some games of Connect 4 with a friend
- best strategy to win?

Connect 4 Strategy

Discussion

Who can force a win?

Connect 4 Strategy

Discussion

Who can force a win?

Discussion

What do you think is the best first move? What is the best second move? What would a complete winning strategy description look like?

Weakly and strongly solved games

Definition

A game is

Weakly and strongly solved games

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A game is

- **very weakly solved** if we know which player can force a win at the start of the game

Weakly and strongly solved games

Definition

A game is

- **very weakly solved** if we know which player can force a win at the start of the game
- **weakly solved** if we know *how* the player can force a win starting from the start of the game

Weakly and strongly solved games

Definition

A game is

- **very weakly solved** if we know which player can force a win at the start of the game
- **weakly solved** if we know *how* the player can force a win starting from the start of the game
- **strongly solved** if we know which player will win and how from any possible state of the game

Examples

Examples

Example 1

Chomp is very weakly solved.

Examples

Example 1

Chomp is very weakly solved.

Example 2

Checkers is weakly solved.

Examples

Example 1

Chomp is very weakly solved.

Example 2

Checkers is weakly solved.

Example 3

Connect 4 is strongly solved (89.6 GB data).

Weak solution

See weak solution on YouTube.
Play against weak solution on Github.

Final thoughts

Reflection

Reflection

- What is the difference between a weak solution and a strong solution of a game?

Reflection

- What is the difference between a weak solution and a strong solution of a game?
- How was a graph used to visualize a weak solution of Connect 4?

Submit your responses by **Tonight!**